

EE 446 Assignment 1
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Problem 1e:

I used the `quantize_and_evaluate` function to convert the distilled student model to int8, float16, and dynamic range. All of these ended up having the same accuracy as the float32 student model, but the float16 and int8 models did end up smaller. The int8 student model was the smallest at 3.95 KB and an accuracy of 0.9630.

Problem 2:

a)

Supervised - Classification	Keras
Transfer Learning (Images)	MobileNetV1
Supervised - Regression	RNN
Unsupervised - Visual anomaly detection	K-means
Unsupervised - Anomaly Detection	GNN

b) Dense - Maps features to class outputs via weighted scores

1D Convolution/pooling - isolates features in 1 dimensional signals (time-sequence)

2D Convolution/pooling - isolates features in 2 dimensional signals (images)

Reshape - changes tensor dimensions so they can be fed into layers of certain shapes

Flatten - converts a multi-dimensional tensor into a 1D vector to be fed into Dense layer

Dropout - reduces overfitting by setting random activations to zero during training

c) Quantized (int8), unoptimized (float32)

d) Black Forest Labs Flux-Pro Synthetic Image Generator

- Uses replicate.com's API

GPT-Image-2 Synthetic Image Generator

- Uses OpenAI's GPT-Image-2 model

Whisper Synthetic Voice Generator

- Uses OpenAI's text-to-speech model

Synthetic data is useful when real data is challenging/expensive to collect and or label. It also allows for expansion and increased diversity of real data, so models can be trained to be more robust.